

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	97.43%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	91.15%	Share of actual attacks successfully flagged
F1-Score	94.19%	Balanced measure combining Precision and Recall
Accuracy	96.25%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	8.85%	Actual attacks that went undetected
True Positive Rate (TPR)	91.15%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.80%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4746	FN 461
Actual: Normal	FP 125	TN 10281

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4746	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	97.43%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	91.15%	Share of actual attacks successfully flagged
F1-Score	94.19%	Balanced measure combining Precision and Recall
Accuracy	96.25%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	8.85%	Actual attacks that went undetected
True Positive Rate (TPR)	91.15%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.80%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4746	FN 461
Actual: Normal	FP 125	TN 10281

2d. Response Latency

Metric	Value	Description
Detection Time	275.9 ms	Average time taken to detect an attack
Mitigation Response Time	6.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.07%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	10.38%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	9.96%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	3	3	2026-09-14 12:00:05	2026-09-14 12:01:27
10.0.0.16	2	2	2026-09-14 12:00:02	2026-09-14 12:01:13
10.0.0.18	2	2	2026-09-14 12:00:06	2026-09-14 12:01:17
10.0.0.19	2	2	2026-09-14 12:00:03	2026-09-14 12:01:14
10.0.0.3	4	1	2026-09-14 12:01:10	2026-09-14 12:01:51
10.0.0.10	1	1	2026-09-14 12:02:14	2026-09-14 12:02:24
10.0.0.6	1	1	2026-09-14 12:01:22	2026-09-14 12:01:32

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-14 12:00:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:00:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:00:05	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:00:06	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:00:13	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-14 12:00:13	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-14 12:00:14	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-14 12:00:14	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-14 12:00:16	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-14 12:00:16	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-14 12:00:17	10.0.0.18	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-14 12:00:17	10.0.0.18	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-14 12:01:09	10.0.0.8	DDoS	UDP Flood	85.2%	High	Time Ban (1m)
2026-09-14 12:01:13	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-14 12:01:14	10.0.0.19	DDoS	SYN Flood	93.0%	High	Time Ban (2m)
2026-09-14 12:01:16	10.0.0.17	DDoS	SYN Flood	94.5%	Critical	Released
2026-09-14 12:01:17	10.0.0.7	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-14 12:01:17	10.0.0.18	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)
2026-09-14 12:01:19	10.0.0.10	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-14 12:01:20	10.0.0.3	DDoS	Uncertain	58.9%	Low	Released
2026-09-14 12:01:27	10.0.0.17	DDoS	SYN Flood	94.5%	Critical	Time Ban (2m)
2026-09-14 12:01:32	10.0.0.6	DDoS	Uncertain	68.7%	Low	Released
2026-09-14 12:02:24	10.0.0.10	DDoS	UDP Flood	73.3%	Medium	Released
2026-09-14 12:02:36	10.0.0.6	DDoS	UDP Flood	92.0%	High	Time Ban (1m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.10	UDP Flood	0.6107	73.3%	Medium	Phase 1	1	10s	2026-09-14 12:02:24	Attack Stopped
10.0.0.3	Uncertain	0.6120	59.9%	Low	Phase 1	1	10s	2026-09-14 12:01:51	Attack Stopped
10.0.0.3	Uncertain	0.6125	59.9%	Low	Phase 1	1	11s	2026-09-14 12:01:41	Attack Stopped
10.0.0.6	Uncertain	0.6197	68.7%	Low	Phase 1	1	10s	2026-09-14 12:01:32	Attack Stopped
10.0.0.3	Uncertain	0.6197	59.9%	Low	Phase 1	1	10s	2026-09-14 12:01:30	Attack Stopped
10.0.0.17	SYN Flood	0.6430	94.5%	Critical	Phase 2	3	11s	2026-09-14 12:01:27	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.3	Uncertain	0.6126	58.9%	Low	Phase 1	1	10s	2026-09-14 12:01:20	Attack Stopped
10.0.0.18	SYN Flood	0.6417	92.5%	Critical	Phase 2	2	1m 11s	2026-09-14 12:01:17	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6088	94.5%	Critical	Phase 2	1	1m 11s	2026-09-14 12:01:16	Ban Expired
10.0.0.19	SYN Flood	0.6013	93.0%	High	Phase 2	2	1m 11s	2026-09-14 12:01:14	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6279	92.9%	Critical	Phase 2	2	1m 11s	2026-09-14 12:01:13	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6462	92.5%	Critical	Phase 2	1	11s	2026-09-14 12:00:17	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6393	93.4%	Critical	Phase 2	1	11s	2026-09-14 12:00:16	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6482	93.0%	Critical	Phase 2	1	11s	2026-09-14 12:00:14	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6376	92.9%	Critical	Phase 2	1	11s	2026-09-14 12:00:13	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE